MWALIMU EXAMINATIONS BUREAU

UACE RESOURCE PRE-MOCK EXAMINATIONS 2019

P525/2 CHEMISTRY PAPER 2 GUIDE

1. (a) (i) Magnesium burns with a bright white flame forming a white solid/powder **(02½ Mks)**

$Mg_{(s)} + H_2O_{(g} \longrightarrow MgO_{(s)} + H_{2\,(g)}$ reject $H_2O_{(l)}$

(ii) Calcium sinks forming bubbles of acolourless gas and a milky solution **(02½ Mks)**

$Ca_{(s)} + H_2O_{(l)} \longrightarrow Ca(OH)_{2\,(s/aq)} + H_{2\,(g)}$

(b)- Beryllium reacts on warming with dilute sulphuric acid to form beryllium sulphate and water(ignore without warming) **(07 Mks)**

$Be_{(s)} + H_2SO_{4\,(aq)} \longrightarrow BeSO_{4\,(aq)} + H_{2\,(g)}$

- Barium reacts very slowly with dilute sulphuric acid and the reactions stops after a short time due to formation of an insoluble barium sulphate. Hydrogen is also evolved.

  $Ba_{(s)} + H_2SO_{4\,(aq)} \longrightarrow BaSO_{4\,(aq)} + H_{2\,(g)}$

- Hot concentrated suphuric acid oxidizes beryllium and barium to their corresponding sulphates and itself reduced to sulphuridioxide and water.

  $X_{(s)} + 2H_2SO_{4\,(aq)} \longrightarrow XSO_{4\,(aq/s)} + H_2O_{(l)} + SO_{2\,(g)}$

The reaction with barium is very slow and takes a very short time.

(c) (i) Solubility decreases from beryllium sulphate to barium sulphate. **(01 Mk)**

The radius of the cations increases form beryllium ion to barium ion as the charge on the cation remains the same. Therefore both the lattice and hydrogen energies decrease from beryllium sulphate to barium sulphate. However the decrease in hydrogen energy is more than that of lattice. This makes the heat of solution of the sulphates to become more endothermic from beryllium sulphate to barium sulphate and hence the solubility decreases in the same order

(03 Mks)

(d) (i) $Ca_{(s)} + 2H_2O_{(l)} \longrightarrow Ca(OH)_{2\,(s/aq)} + C_2H_{2\,(g)}$ **(1½ Mks)**

(ii) $Be_{(s)} + 2OH^-_{(aq)} + 2H_2O_{(l)} \longrightarrow Be(OH)^{2-}_{4\,(aq)} + H_{2\,(g)}$ **(1½ Mks)**

(e) - Beryllium atom has the <u>smallest</u> atomic radius **(01 Mk)**

- Beryllium cation has the smallest ionic radius
- Beryllium cation has the highest charge density/polarizing power
- Beryllium atom has the highest electronegativity/lowest electropositivity

NB: *Any one correct ignore extra wrong/ correct reason*

2. (a) Is the number of times (a naturally occurring) atom of an element is heavier than a twelfth of the mass of an atom of a carbon-12 isotope. **(01 Mk)**

(b) (i) The apparatus is evacuated of air. Thee vapourised sample of the element is introduced into the ionizing chamber where it collides with fast moving electrons to form positive ions. The positive ions formed are accelerated by a strong electric field so that the ions which enter the magnetic field will be of the same kinetic energy. The magnetic field deflects the ions according to the mass to charge ratio on the ion collector.

The strength of both the magnetic and electric fields are varied so that ions of a given mass are received at the detector. The collected charges are <u>amplified</u> into sizeable currents which operates a pen and draws the mass spectrum of the element being investigated (The collected charges are <u>amplified</u> and finally <u>recorded</u>) **(08 Mks)**

(ii) Let % of 63_{Cu} be x **(03 Mks)**

$$\therefore \%\text{ of } 65_{Cu} = (100 - x)\text{ RAM} = \sum(\text{isotopic mass} \times \%\text{ abundance})$$

$$65.55 = \left(\frac{x}{100} \times 63\right) + \left(\frac{100 - x}{100} \times 65\right)$$

$$63.55 = 63x + 6500 - 65x$$

$$x = 72.5$$

% abundance of 65_{Cu} is 72.5 and 65_{Cu} is 27.5

(iii) Chlorine has isotopes 35_{Cl} and $37_{Cl.}$ The peaks recorded at mass member 70, 72 and 74 were due to the following molecular ions **(02½ Mks)**

$[35_{Cl} - 35_{Cl}]^+, [35_{Cl} - 37_{Cl}]^+$ and $[37_{Cl} - 37_{Cl}]$ respectively

(iv) – Fast **01 Mk)**

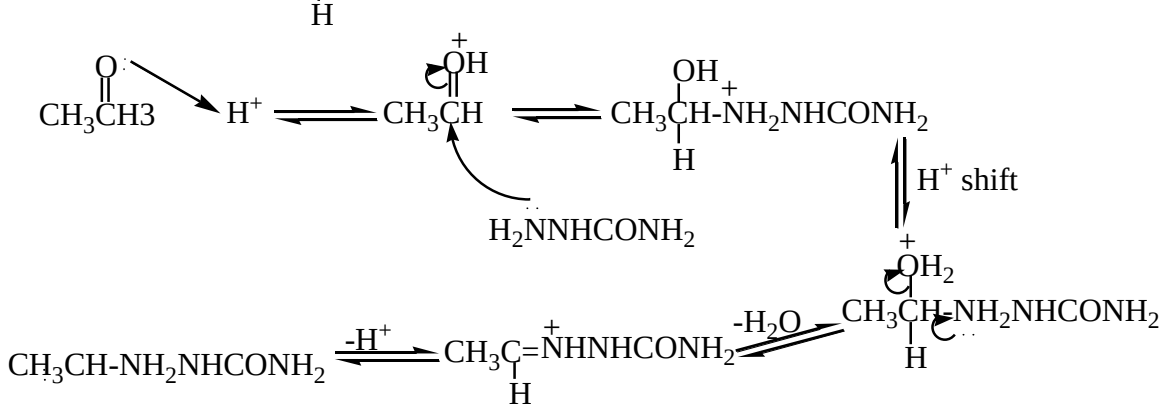
- Accurate
- Requires minute sample for analysis

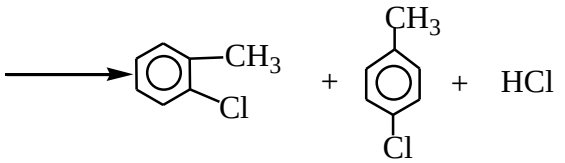
(c) (i) $X^+_{(g)} \longrightarrow X^{2+}_{(g)} + e^-$ (0½ for no/wrong states) **(01 Mk)**

(ii) Ionization energy increases from first to forth. This is because as successive electrons are removed, the number of protons becomes greater than the remaining electrons and this causes remaining electrons experience a stronger nuclear attraction and this causes the ionization energy to successively increase. There is <u>a greater increase in ionization energy from second to third compared to from first to second and third to fourth</u> because the third electron is being removed an inner freely energy level which is freely filled and stable and nearer to the nucleus. Thus the electrons experiences stronger nuclear attraction/change and hence greater energy is needed to remove the electron. **(3½ Mks)**

(05 Mks)

3. (a) $\longrightarrow CH_3C(H)=NNHCONH_2$ (+ H_2O)



(b) 

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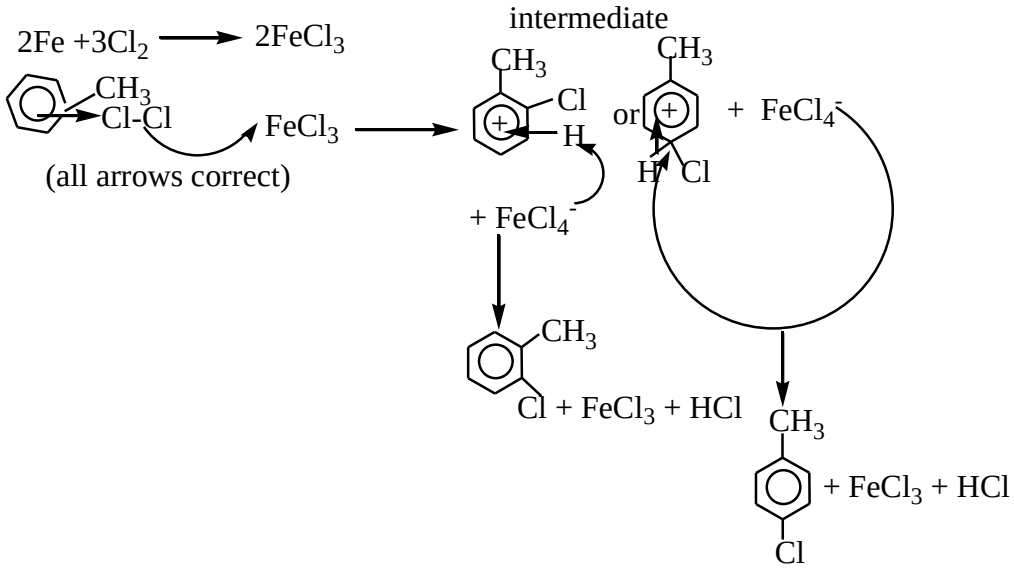
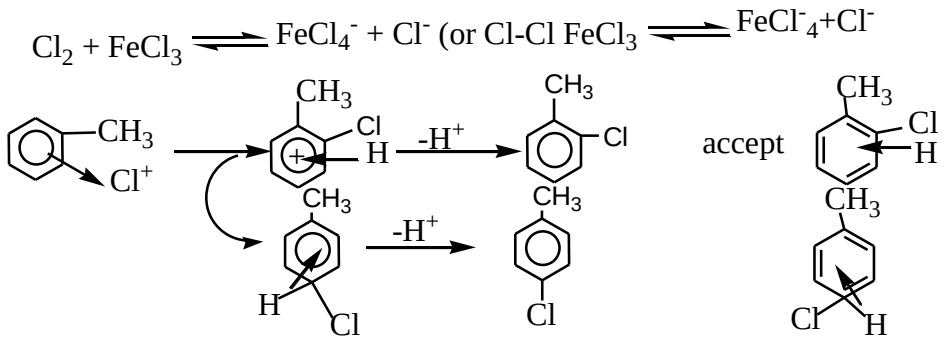
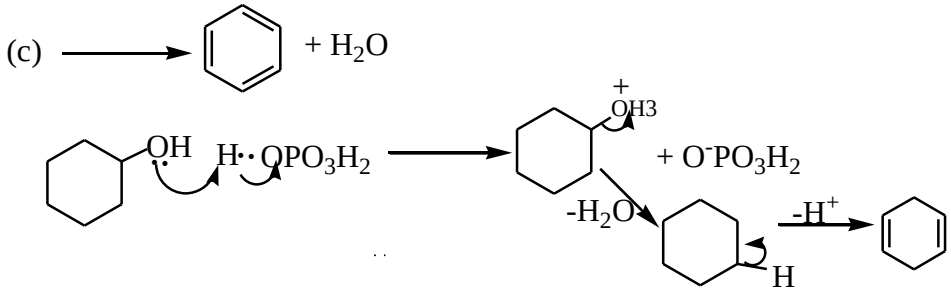


Alternatively;



(c) 

(03$^{1/}_{2}$ Mks)

4. (a) (i) Is the heat/enthalpy change when one mole of hydrated hydrogen ions form an acid reacts with one mole of hydrated hydroxide ions from an alkali to form one molecule of water **(01 Mk)**

(ii) Enthalpy change when one mole of an ionic salt dissolves in a large excess of water (solvent) to form an infinitely dilute solution. **(01 Mk)**

(b) – A known volume (V_1cm^3) of hydrochloric acid of known molarity (a $moldm^{-3}$) is placed in a plastic cup/beaker (or well insulated calorimeter) ***No marks indicated***

- The initial temperature of the acid s noted and recorded as $t_1{}^oC$.
- An <u>equal volume</u> of sodium hydroxide of an <u>equal molarity</u> (or slightly higher molarity) is measured and its initial temperature noted and recorded as $t_1{}^oC$.
- The alkali is quickly added to the acid in the cup/beaker/calorimeter, mixtures turned well using a thermometer and the <u>highest temperature of the resultant solution noted and recorded as $t_3{}^oC$.</u>

Results

- Temperature change, $\Delta t\left(t_3\left(\frac{t_1+t_2}{2}\right)\right)°C$
- Total where of solution $=2V_1cm^3$
- Assume the solution is sufficiently dilute, density of solution = $1gcm^3$ and specific heat capacity of solution $= 4.2Jg^{-1}°C^{-1}$.
- Mass of solution $= (2V_1\times1) = 2V_1g$
- Assuming no heat losses to the surroundings. Heat evolved $= 2V1\times 4.2 \times \Delta t = (8.4V_1\Delta t)J$

No of moles of acid that reacted $= \frac{aV_1}{1000}$

No of moles of alkali reacted $= \frac{aV_1}{1000}$

No of moles of H_2O formed $= \frac{aV_1}{1000}$

$= \frac{aV_1}{1000}$ moles of H_2O evolve $(8.4V_1\Delta t)$ J of heat

Enthalpy of neutralization $= -\left(\frac{8.4V_{1\Delta t}}{\frac{aV_1}{1000}}\right)Jmol^{-1}$

or $-\left(\frac{8.4\Delta t}{a}\right)KJmol^{-1}$

(c) Axes shape plotting **(03 Mks)**

(d) (i) If a system A is converted into a system B the heat absorbed or evolved is independent of the method of passing from A to B. **(01 Mk)**

(ii) I. $C_2H_6 + \frac{7}{2}O_{2(g)} \longrightarrow 2CO_{2(g)} + 3H_2O_{(l)}\ ;\Delta H_1 = -1542$ **(04 Mks)**

II. $C_2H_{2(g)} + \frac{5}{2}O_{2(g)} \longrightarrow 2CO_2 + H_2O_{(l)}\ ;\Delta H_2 = -1310$

(III). $H_{2(g)} + O_{2(g)} \longrightarrow H_2O_{(l)}\ ;\Delta H_3 = -285$

IV. $C_2H_{2(g)} + 2H_{2(g)} \longrightarrow C_2H_{6(g)}\ ;\ \Delta H_4 = x$

$x = \text{eqn II.} + 2\text{eqn III} + \text{reserved I}$

$= (-1310) + 2(-285) + (+1542)$

$= -338KJmol^{-1}$

5. (a) P; $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$ S; $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$

In phosphorus, the election is added to a <u>half filled 3p-sub-energy level</u> which is energetically stable. Therefore the incoming electron experiences more repulsion than nuclear attraction. Hence lower electron affinity whereas in sulphur, the electron is added to <u>3p sub-energy level with four electrons which is unstable</u>. The electron experiences more nuclear attraction than repulsion hence higher electron affinity. **(03½ Mks)**

(b) Both sodium ion and rubidium ion have the same charge but sodium ion has a smaller ionic radius than rubidium ion. This makes the sodium ion to have a higher charge density than rubidium ion. Sodium ion is therefore heavily hydrated in aqueous sodium than rubidium ion.

Effective size of sodium ion increases and its ionic mobility decreases. Rubidium ion is less heavily hydrated and this makes it more mobile in aqueous solution. **(03½ Mks)**

(c) Molecules of ethanoic acid posses a highly polar oxygen-hydrogen bond due to the oxygen atom being highly electronegative. These molecules of ethanoic acid are held by hydrogen bonds while molecules of methylmethanate do not have a highly polar oxygen-hydrogen bond (are non-polar). Thus their molecules are held by Van der Waals forces of attraction. The hydrogen bonds are stronger than the hydrogen bonds. **(03½ Mks)**

(d) Hot concentrated sulphuric acid oxidizes hydrogen iodide to iodine (seen as purple vapours) and itself reduced water and sulphur dioxides. **(04 Mks)**

$$2HI_{(g)} + H_2SO_{4\,(aq)} \longrightarrow H_2O_{(l)} + SO_{2\,(g)} + I_{2\,(s)}$$

Hydrogen chloride is less reducing than hydrogen iodide and thus <u>cannot be oxidized</u> by hot concentrated sulphuric acid to chlorine

(f) The chromium (III) ions have a small ionic radius and high charge and have high charge density. Chromium (III) ions are heavily hydrated and undergo hydrolysis to release hydrogen ions which makes the solution acidic. **(05½ Mks)**

The acidic solution reacts with magnesium ribbon to form hydrogen (seen as bubbles of a colourless gas) and insoluble <u>chromium (III) hydroxide</u> seen as a green precipitate.

$$2Cr^{3+}_{(aq)} + 3Mg_{(s)} + 6H_2O_{(l)} \longrightarrow 2Cr(OH)_{3(s)} + 3Mg^{2+}_{(aq)} + 3H_{2(g)}$$

or $2Cr(H_2O)^{3+}_{6(aq)} + 3Mg_{(s)} \longrightarrow 2Cr(H_2O)_3(OH)_{3(s)} + 3Mg^{2+}_{(aq)} + 3H_{2(g)}$

6. (a) Sodium metal **(02 Mks)**

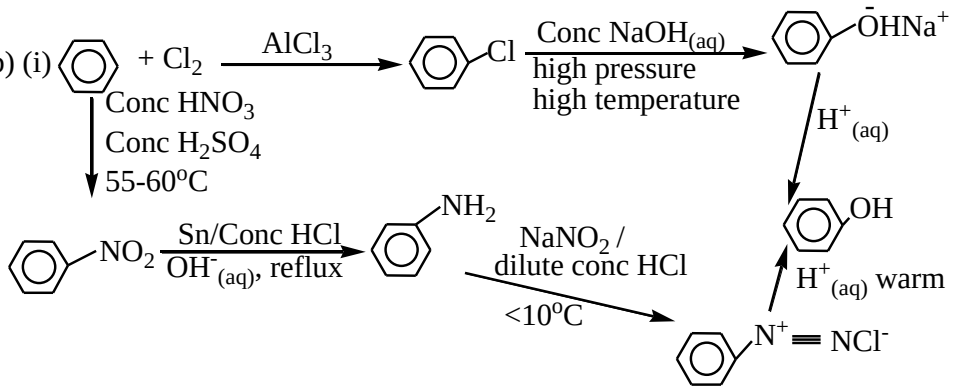
Y and Z give bubbles of a colourless gas

(ii) Neutral iron (III) chlorides solution **($2^{1}/_{2}$ Mks)**

Y; Purple/ nolet colouration

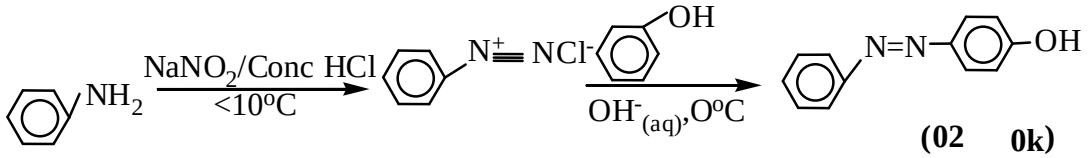
Z; No observation change

(b) (i)



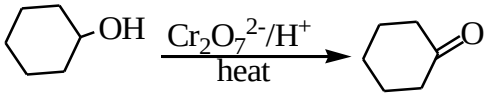
(03 Mks)

(ii)



(02 0k)

(c) (i)



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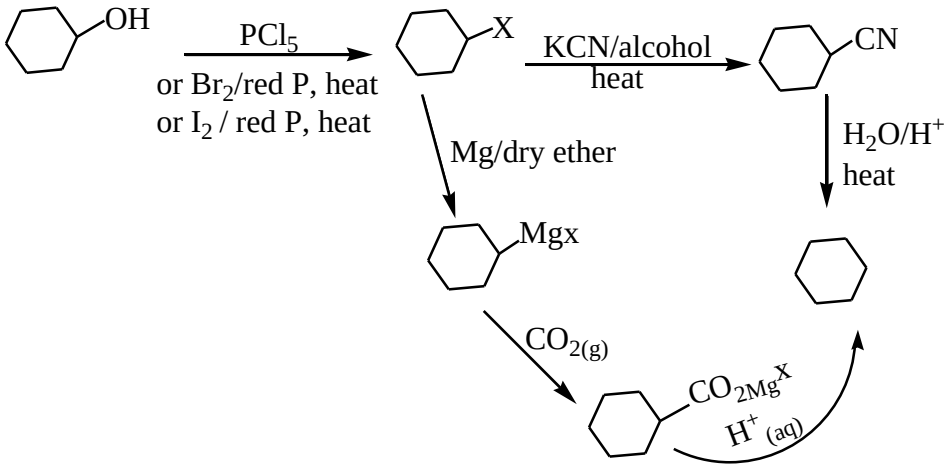
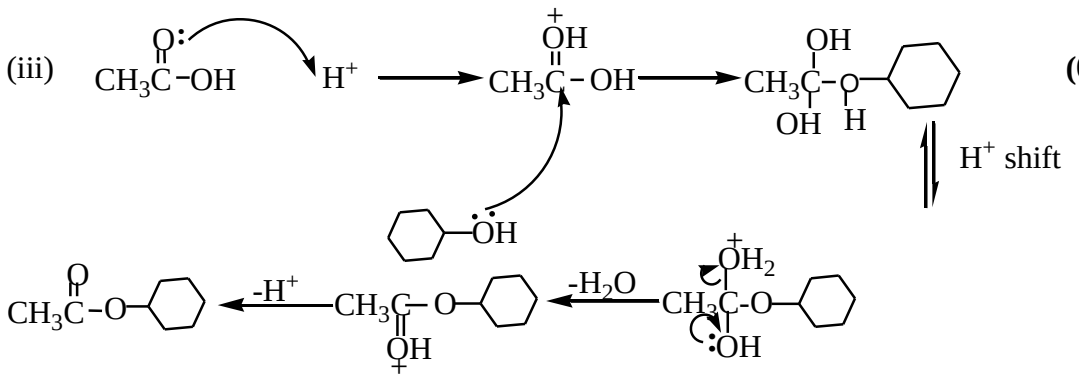


(02 Mks)

(iii) 

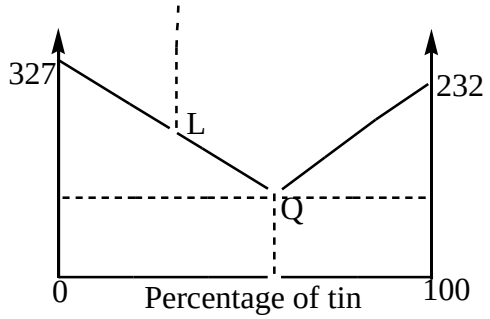
(04 Mks)

(d) The lone pairs of electrons on the oxygen atom in Y interact with the pi electrons of the benzene ring. This makes the oxygen-carbon bond stronger while the oxygen-hydrogen bond weakens. As a result Y loses a proton easily in aqueous solution which makes it acid. Whereas in Z the intervening methylene group prevents such interaction that occurs in Y. Hence the oxygen-hydrogen bond is not weakened and the proton is <u>not easily lost</u> in aqueous solution. **(03 Mks)**

7. (a) (i) A liquid mixture which at constant pressure solidifies at constant temperature to give a solid mixture of the same composition as the liquid mixture **(01 Mk)**

Or A solid mixture which at constant pressure melts at constant temperature to give a liquid mixture of the same composition as the solid mixture.

(ii) Is the constant temperature at which a liquid mixture of two components is in equilibrium with its solid components at a given pressure. **(01Mk)**

(b) (i) 

(06 Mks)

(drawing)

(ii) Eutectic temperature = $183^{o}C \pm 2$ **(02 Mks)**

Eutectic composition is 62% $\pm$ 1 tin and 38% $\pm$ 1 lead

(iii) The liquid mixture cools up to point L without change in phase. At L, pure solid lead begins to crystallize out. Composition of tin increases in the mixture and this causes a further decrease in freezing point along LQ on the curve. (with further cooling, more solid lead crystallizes leaving a liquid mixture richer in tin than lead. The composition of the liquid tends towards point Z). **(05 Mks)**

At Q, solid tin also begins to crystallize out together with solid lead. The freezing temperature and composition remain constant until all the liquid mixture freezes.
Thereafter, the solid mixture cools further without change in phase up to 100°C.

(c) (i) – Changing pressure causes the composition of the mixture to change **(03 Mks)**

- X-ray diffraction pattern that does not conform to that of a pure compound is obtained
- Microscopic examination shows a heterogeneous mixture of individual crystals of the components that make up the eutectics mixture.

(ii) Soldering rej. Solder

(d) Zinc-cadmium, Antimony-lead, cold-thallium Bismuth cadmium (must be a pair) **(01 Mk)**

8. (a) ZnS **(01 Mk)**

(b) (i) -The ore is ground to power and added to a mixture of water and an oil/frothing agent.

- The mixture is agitated by blowing in air
- The ore particles (and frothing agent) are carried to the surface as a froth which is removed/ skimmed off. An acid is added to it, mixture is filtrated off to obtain the concentrated ore which is then dried.
- Impurities sink to the bottom. **(03½ Mks)**

(ii) $2ZnS_{(s)} + 3O_{2\,(g)} \longrightarrow 2ZnO_{(s)} + 2SO_{2\,(g)}$ Reject $ZnCO_3$ if used **(1½ Mks)**

$2PbS_{(s)} + 3O_{2\,(g)} \longrightarrow 2PbO_{(s)} + 2SO_{2\,(g)}$ (ignore)

(iii) The mixture of zinc oxide, limestone and coke are put in a blast furnace and heated by stream of hot air in the furnace, zinc oxide is reduced by carbon monoxide formed or coke to zinc (vapour) **(04½ Mks)**

$ZnO_{(s)} + CO_{(g)} \longrightarrow Zn_{(g)} + CO_{2\,(g)}$

$ZnO_{(s)} + CO_{(g)} \longrightarrow Zn_{(g)} + CO_{(g)}$

The zinc vapour distills off from the furnace and is cooled by spray of molten lead. A zinc form is purified by redistillation.

(c) (i) Brown/yellow/ reddish brown solution turns green **(02½ Mks)**

$2Fe^{2+}_{(aq)} + Zn_{(s)} \longrightarrow Zn^{2+}_{(aq)} + 2Fe^{2+}_{(aq)}$

(ii) White precipitate soluble/dissolves to form a colourless solution **(04 Mks)**

$Zn^{2+}_{(aq)} + 2OH^{-}_{(aq)} \longrightarrow Zn(OH)_{2(s)}$

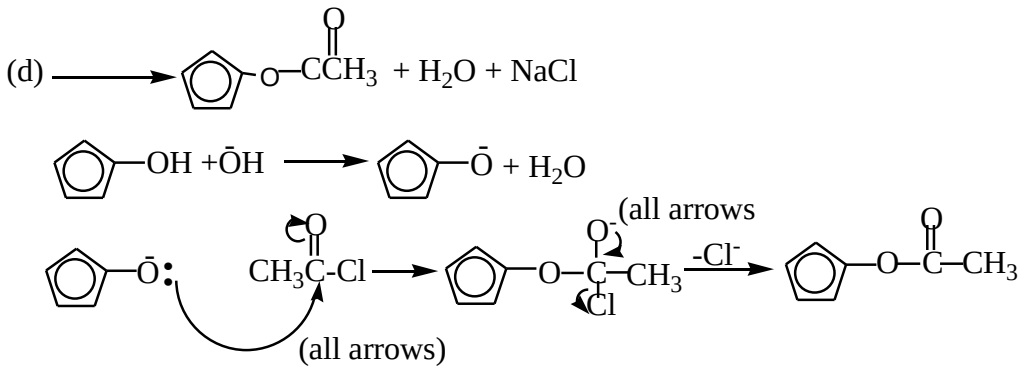
$Zn(OH)_{2(s)} + 4NH_{3\,(aq)} \longrightarrow Zn(NH_3)^{2+}_{4\,(aq)} + 2OH^{-}_{(aq)}$

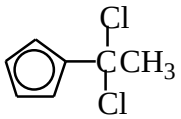
(iii) Grey solid dissolves with effervescence/bubbles of a colourless gas forming a colourless solution **(03 Mks)**

$Zn_{(s)} + 2OH^{-}_{(aq)} + 2H_2H_{(aq)} \longrightarrow Zn(OH)^{2-}_{4(aq)} + H_{2(g)}$

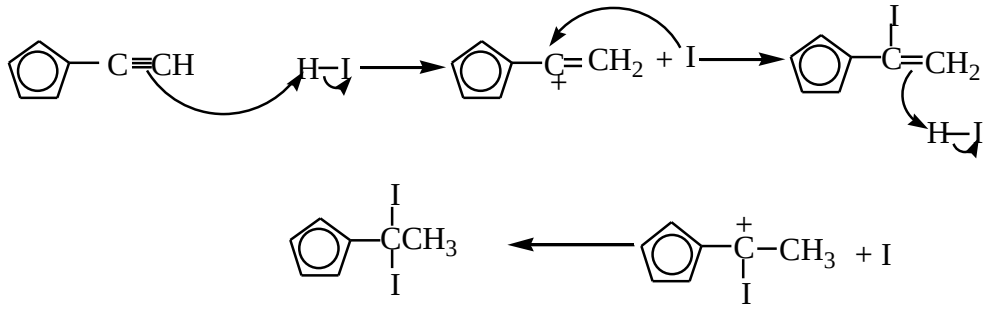
or $Zn_{(s)} + 2OH^{-}_{(aq)} \longrightarrow ZnO^{2-}_{2(aq)} + H_{2(g)}$

(d)



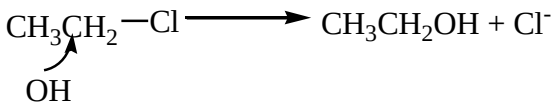
(e) 

(2 $^1/_2$ Mks)



(f) $CH_3CH_2OH + Cl^-$

(02 Mks)



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